Academic Advising Assistant Using Retrieval-Augmented Generation

Abstract

This project develops an Academic Advising Assistant using Python, LangChain, Retrieval-Augmented Generation (RAG), and a large language model deployed through a Streamlit application. The objective is to enable students to ask academic advising questions and obtain accurate information from institutional documents such as degree plans, course catalogs, and class schedules. The key outcome is that the system retrieves relevant document segments from a vector database and generates responses using a locally hosted LLM through the Ollama framework.

Introduction

Academic advising supports students in understanding degree requirements, selecting courses, and planning their academic progress. Students frequently ask similar questions regarding prerequisites, degree plans, and course schedules, making automated assistance useful for improving access to information.

Retrieval-Augmented Generation (RAG) enables large language models to answer questions using external documents by retrieving relevant information and providing it as context. The goal of this project is to develop a RAG-based academic advising assistant that answers student questions using institutional advising documents. The system integrates Python, LangChain, vector embeddings, and a locally hosted LLM within a Streamlit interface to provide an interactive advising tool.

Methods

Data Source: The system uses institutional academic advising documents containing information on degree requirements, course descriptions, prerequisites, instructor assignments, and class schedules.

Document Processing: Documents were loaded and processed using Python and LangChain. Word, PDF, and Excel files were converted into text format and split into smaller chunks using a Recursive Character Text Splitter to improve retrieval performance.

Embeddings and Vector Database: Text chunks were converted into vector embeddings using the mxbai-embed-large embedding model through Ollama. The embeddings were stored in a Chroma vector database, enabling semantic search to identify relevant document sections for user queries.

Retrieval-Augmented Generation (RAG): The system uses a RAG pipeline in which user questions are matched with relevant document chunks retrieved from the vector database. The retrieved context is combined with the user query in a prompt template, and a locally hosted LLM via Ollama generates the final response based on the retrieved information. The system design prioritizes grounded responses by restricting the language model to information retrieved from institutional documents, reducing the likelihood of hallucinated answers.

Application Interface: The system is deployed using Streamlit, which provides an interactive chat interface where users can submit advising questions and receive responses generated by the RAG system.

Output Examples

Figures 1-7 present example interactions between users and the academic advising assistant. These examples demonstrate the system's ability to answer questions related to course requirements, 1

Team 4 prerequisites, class schedules, and instructor information based on retrieved institutional documents.

What are the core courses in Marketing?

MKTG 3100 Marketing Research MKTG 4050 Applied Marketing Management

Source:

- marketing_4yp_2024-25.docx

What are the graduation requirements for Accounting?

Here is the answer to your question:

- Complete a minimum of 120 applicable course hours.
- Achieve a minimum 2.0 CU cumulative GPA and business GPA.
- Complete a minimum of 45 upper division (3000/4000 level) credits.
- Complete a minimum of 30 CU Denver business course hours in good standing.

Note: These requirements are specifically for Accounting majors, as indicated by the "CU Denver Business students" phrase.

Source:

- accounting_4yp_2024-25.docx

What is the course information for BANA 6620?

Computing for Business Analytics (3 Credits)

Introduces database and modeling software used by business analytics professionals. Includes querying relational databases, state-of-the-art statistical freeware, and modeling software. Students learn to obtain, organize, and store data needed for analytics projects, undertake data cleansing for big data tasks, and conduct statistical data visualization.

Restriction: Restricted to BANA-MS students within the Business School. Max Hours: 3 Credits. Grading Basis: Letter Grade

Source:

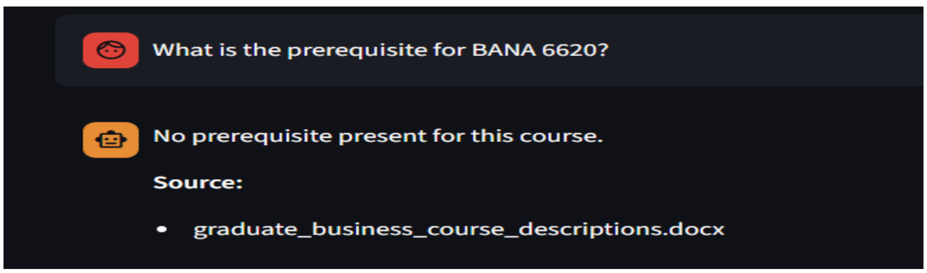
- graduate_business_course_descriptions.docx

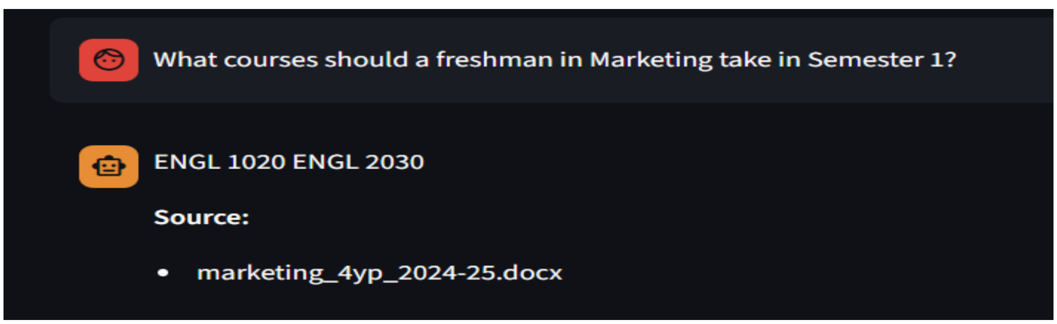
What are the time and location details for BANA 6620 in Fall 2025?

Start Date: 2025-08-18 End Date: 2025-10-11 Start Time: 18:30 End Time: 21:14:59.999000 Location: DC Building: D_BUSB Room: 3100

Source:

- Fall 2025 Class List_ZW.xlsx





Conclusion

The RAG-based academic advising assistant successfully retrieves relevant information from institutional documents and generates responses to common student advising questions. Overall, the system performs well in answering queries related to course requirements, prerequisites, schedules, and instructor information.

However, some limitations were observed. Converting structured tables from spreadsheets into text can sometimes reduce formatting clarity, which may affect response accuracy. In addition, system performance depends on effective document chunking and retrieval quality.

Future improvements could include better handling of tabular data, storing structured scheduling information in databases, and refining retrieval strategies to further improve response accuracy and reliability.

References

Chroma. (2024). *Chroma vector database.* https://www.trychroma.com

LangChain. (2024). *LangChain documentation.* https://python.langchain.com

Ollama. (2024). *Ollama: Run large language models locally.* https://ollama.ai

Streamlit. (2024). *Streamlit documentation.* https://streamlit.io

Appendices